

Radio Sweden threat attack model with security countermeasures

Group 15

Depending on Lecture slides(4) and what is showing to create an attack model, we decide to through the 3 steps below:

Step1: Decompose the system

In this step, we will obtain to know about the system procedure and external entities operation.

Identify the entry/exit points:

Entry All pcs, Cisco Routers, Cisco ASA firewall, wireless networks, hp office with gigabit switches.

Exit All pcs, backup servers, 7 GNU/Linux Redhat servers for intranet, web servers, email, intrusion detection systems, log, FTP servers.

14 Windows servers 2016 for internal systems (finance, internal communication, systems)

Identify the Assets:

Human: Employees, Freelance journalist

Infrastructure: Firewalls, Routers, Switches, Servers, Laptops,

Systems: Web Server, FTP, Backup servers, payroll system, Anti-malware software, Active Directory server, VPN server,

Data: Produced programs, bought material, employees information

Identify the Assets that may be of interest to different attackers

- Servers
- Radiofrequency
- The identities of employees.
- Secret information

The external dependencies:

They require patches, updates from the external entities below:

Cisco , Microsoft, HP, Solaris, Redhat

Step2 and 3: Determine the threats and their countermeasures

To determine analyze and rank the threats we will use the STRIDE model

Spoofing

Examples: IP address spoofing, ARP spoofing

Explanation: This kind of attack could illegally use Radio Sweden's employee's credentials.

Security Control: Authentication

Countermeasures: be sure before clicking, check the connection, two-factor authentication, using spoofing filter.

Tampering

Examples: Destroying, manipulating, or editing data through the unauthorized medium.

Security Control: integrity

Countermeasures: encryption, data authentication software, monitoring software, and Variable integrity checking.

Repudiation

Examples: removing sensitive data, users info, and logs, privilege rights

Security Control: Non-repudiation

Countermeasures: using digital signatures, anonymous access disabling, the powerful authentication process

Information disclosure

Examples: Extract employee data, Disclosure the produced material, Password disclosure, Unprotected data on the FTP server

Security Control: confidentiality

Countermeasures: Enable multi-factor authentication, perform proper ACL (Access Lever Control), Training the employees that prevent people from accessing the equipment.

Denial of service

Examples: DoS attack against the web server, DoS attack against the FTP

Security Control: availability

Countermeasures: Cloudflare checks and filtering of requests. Provide IDS (Intrusion Detection System) and IPS (Intrusion Prevention System) and use Powerful anti-virus and anti-spyware software in the systems.

Elevation of privilege

Examples: Gain root-level privileges on a server or network equipment, Running the web server as a non-root user.

Security Control: Authorization

Countermeasures: save critical information on the safe side with encryption,